PARTICLE IMPLEMENTATION

by Nour Elmaghalawy

INFO-STEALER LAB SIMULATION

TECHNICAL DETECTION REPORT

12/23/2025

1. Objective

Simulate and detect Info-Stealer malware activity in a controlled lab environment by identifying:

- Privilege escalation behavior
- Malicious PowerShell execution
- Suspicious process creation and chaining

“All activity was simulated using Atomic Red Team techniques. No real malware, credentials, or data exfiltration were involved.”

1.Preparation

Lab Environment

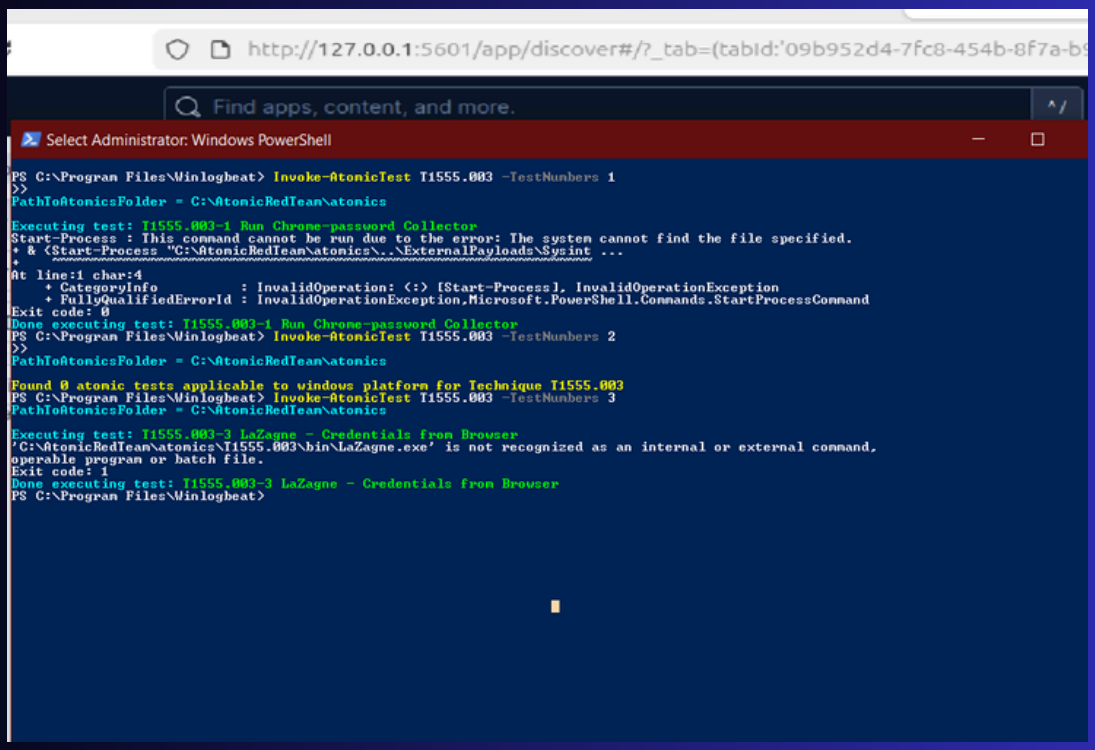
Component Description:

1. Windows Host Target endpoint executing simulated Info-Stealer activity
2. Ubuntu VM Self-hosted Elastic Stack (Elasticsearch, Kibana, SIEM)
3. Log Shipper Winlogbeat (Windows Security Logs to Elastic SIEM)
4. Simulation Tool Atomic Red Team (PowerShell-based tests)

Simulated Attack Scenario

Employee receives a phishing email containing a malicious attachment.
Attachment launches hidden PowerShell with encoded commands (-enc).
PowerShell executes in memory to prepare credential harvesting (simulated).
PowerShell spawns additional processes to simulate payload execution.

[Figure 1 – Atomic Red Team PowerShell Execution on Windows Host]



2. Detection & Analysis

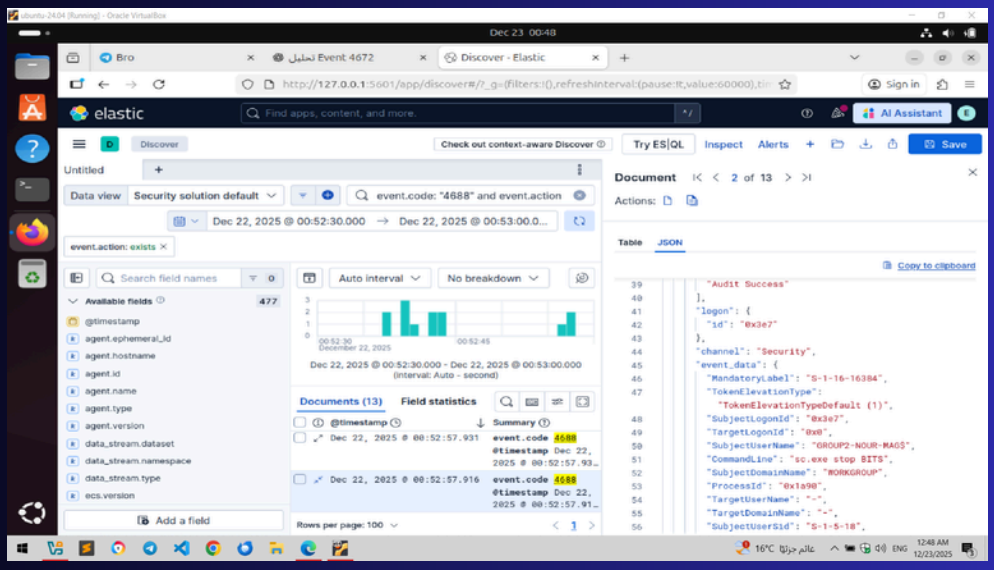
Detection was achieved through analysis of Windows Security Events ID 4688, 4104, 4672 identifying suspicious PowerShell execution spawned during the simulation.

Key indicators included encoded command-line arguments, hidden execution flags, and abnormal parent-child process relationships.

2.1 key Events Summary

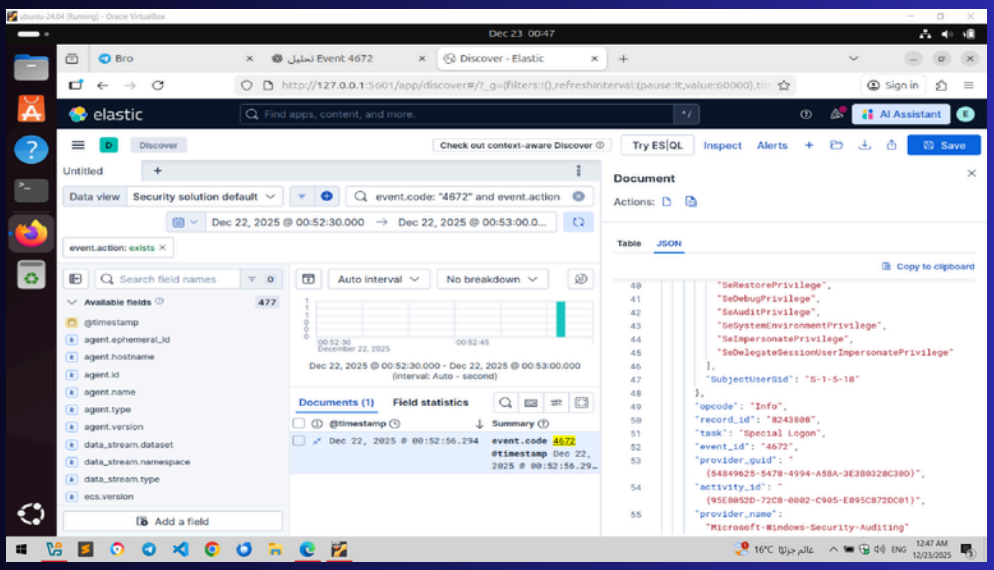
Event ID: 4688
Host: GROUP2-NOUR-MAGHALAWY
User Description: Admin
Description: PowerShell spawning cmd.exe
SOC Relevance: Malicious process chaining

Event 4688 (Process Creation) in Kibana



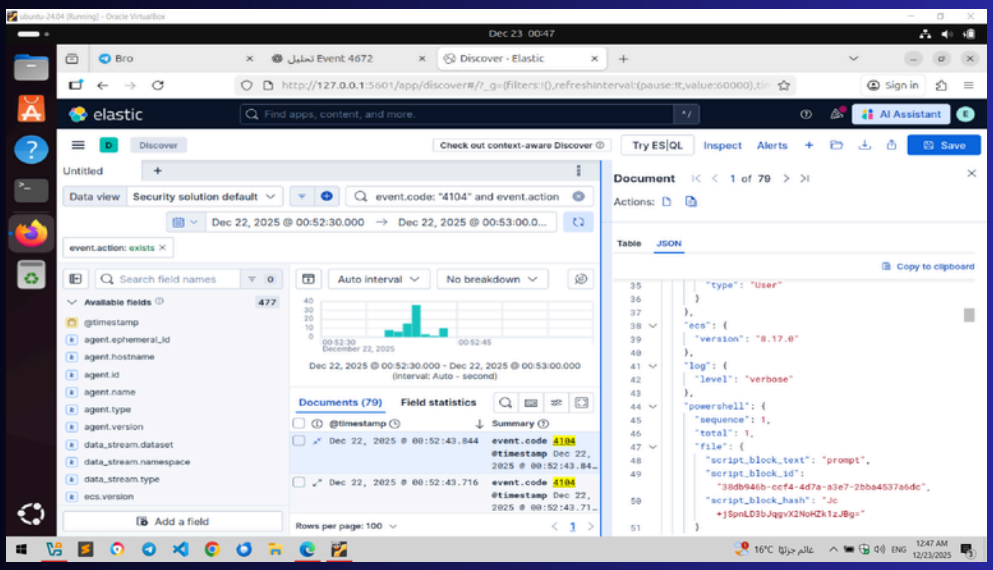
Event 4688 – Suspicious Process Creation
Parent Process: powershell.exe
Child Process: cmd.exe
Technique used: AppPath Bypass
Token Elevation: Full / Elevated
This behavior strongly indicates payload execution or command staging.

Event ID: 4672
Host: GROUP2-NOUR-MAGHALAWY
User: SYSTEM
Description: Special privileges assigned (SeDebug, SeImpersonate, etc.)
SOC Relevance: Indicates elevated privileges



Event 4672 – Privilege Escalation
Account granted special privileges:
SeDebugPrivilege
SeImpersonatePrivilege
SeBackupPrivilege
This event indicates that a high-privilege context is available, which is commonly abused by Info-Stealer malware prior to credential access.

Event ID: 4104
Host: GROUP2-NOUR-MAGHALAWY
User Description: Admin
Description: PowerShell Script Block execution
SOC Relevance: Suspicious PowerShell usage



Event 4104 – PowerShell ScriptBlock Execution
Executed by Admin-level account
ScriptBlock ID and hash captured
Indicates PowerShell executed potentially malicious logic in memory
High-risk when correlated with privilege escalation and process creation events.

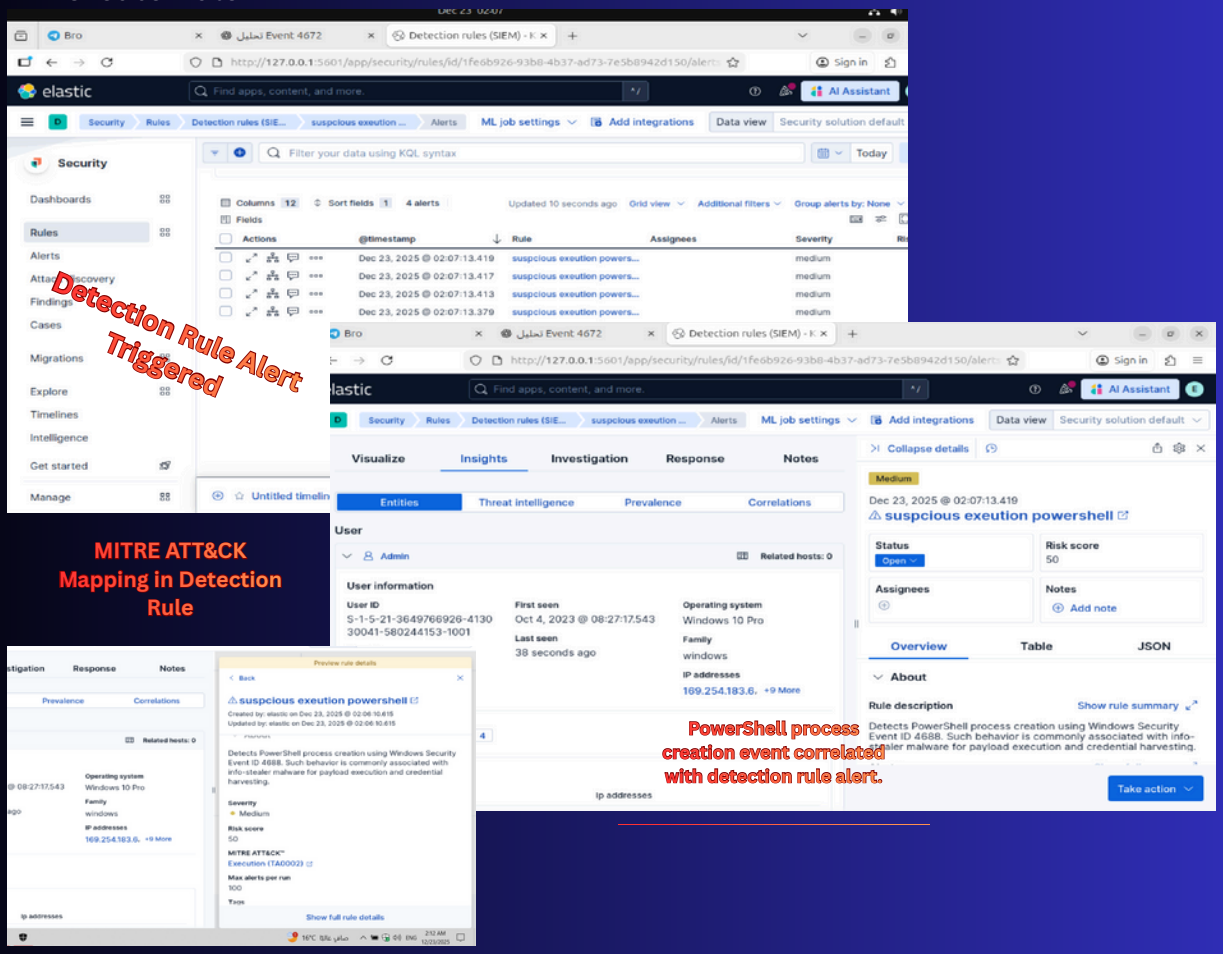
Create Detection Rule in Elastic SIEM

- Go first in Kibana from Security choose Alerts then create new Rule

- Rule Type choose Custom Detection Query

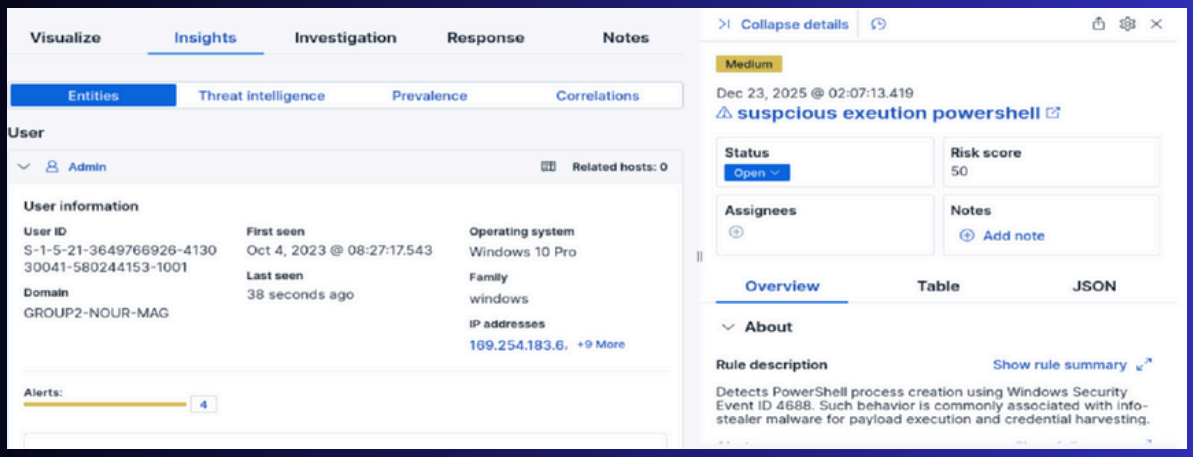
- Define Rule Query:
  event.code:4688 AND process.parent.name:"powershell.exe" AND process.name:"cmd.exe" OR process.name:"powershell.exe")

- Schedule: Every 5 minutes
- Rule Details: Suspicious PowerShell Execution – Info Stealer
- Description: Detects PowerShell process creation using Windows Security Event ID 4688. Such behavior is commonly associated with info-stealer malware for payload execution and credential
- Severity: Medium
- Risk score: 50
- Tags: Windows | PowerShell | Info-Stealer | Atomic-Red-Team
- MITRE ATT&CK Mapping: TA0002 – Execution | T1059.001 – PowerShell | 4104, 4688
- Create Rule



3. Scope
Post-Detection

- **Identify affected hosts**
- **Identify affected user accounts**
- **Determine execution timeframe**
- **Assess potential credential exposure (simulated)**



Host and User Scope in Elastic SIEM

4. Containment

- Endpoint isolation
- Disable suspicious user accounts
- Block outbound connections

Upon detection, the affected endpoint would be isolated from the network, and outbound communication to suspicious destinations would be blocked.

5. Eradication

- Remove malicious scripts and Artifacts
- Clear persistence mechanisms
- Reimage endpoint if required

6. Recovery

- Reset credentials
- Restore normal operations
- Monitor detection rules for recurrence

7. Lessons Learned

- Native Windows logs are sufficient for early Info-Stealer detection
- PowerShell remains a primary attack vector
- Behavioral rules outperform signature-based detection
- improving defense
- editing and create detection rule

This lab showed that defensive capabilities can be improved by creating simple, behavior-based detection rules using Windows Event ID 4688 can effectively identify earlystage InfoStealer activity.
The implemented Elastic SIEM rule detecting PowerShell-initiated process creation served as a pseudo detection use case for early Info-Stealer activity. Although limited to a single attack stage, the rule provided valuable early visibility into malicious behavior. When combined with additional context such as privilege escalation and PowerShell ScriptBlock events, this pseudo detection can be evolved into a high-confidence detection, demonstrating how incremental rule development strengthens overall SOC defense.

The implemented rule, which detects PowerShell spawning suspicious child processes, served as a pseudo detection use case that successfully surfaced malicious execution behavior without relying on malware signatures.
While the rule alone does not confirm a full compromise, it provides a strong foundation for alerting and investigation. Correlating this rule with privilege escalation (Event 4672) and PowerShell ScriptBlock logging (Event 4104) significantly increases detection confidence and reduces false positives.

Detection Logic (Pseudo Logic)

IF
Event ID = 4688
AND Parent Process = powershell.exe
AND Child Process IN (cmd.exe, powershell.exe)
THEN
Flag Suspicious Process Chaining

IF
(4672 AND 4104 AND 4688) occur on same host within short time window (e.g., 10 minutes)
THEN
Raise HIGH-PRIORITY ALERT:
"Potential Info-Stealer Activity – Early Stage"